Ballast Water and Sediments, 2004 (BWM Convention), which aims to minimize the transfer of invasive aquatic species through ships' ballast water and sediments; and .2 the International Convention on the Control of Harmful Anti-fouling Systems on Ships, 2001 (AFS Convention), which addresses anti-fouling systems on ships and focuses on the prevention of adverse impacts from the use of anti-fouling systems and the biocides they may contain.

1.9 Biofouling management practices may also improve a ship's hydrodynamic performance and can be effective at enhancing energy efficiency and reducing air emissions from ships.

This concept has been identified by IMO in the 2022 Guidelines for the development of a ship energy efficiency management plan (SEEMP) (resolution MEPC.346(78)).

These Guidelines further support the 2023 IMO Strategy for the reduction of green house gases from ships (resolution MEPC.377(80)).

- MEPC.346(78)).

These Guidelines further support the 2023 IMO Strategy for the reduction of green house gases from ships (resolution MEPC.377(80)).

1.10 A GEF-UNDP-IMO GloFouling Partnerships Project was conducted as part of wider efforts by IMO, in collaboration with the United Nations Development Programme (UNDP) and the Global Environment Facility (GEF), to protect marine ecosystems from the negative effects of invasive aquatic species.

The aim of the GloFouling Partnerships Project was to build capacity in developing countries for implementing the IMO Biofouling Guidelines and other relevant guidelines to minimize the transboundary introduction of invasive aquatic species, with additional benefits in the reduction of greenhouse gas emissions from global shipping.

RESOLUTION MEPC.378(80) (adopted on 7 July 2023) 2023 GUIDELINES FOR THE CONTROL AND MANAGEMENT OF SHIPS' BIOFOULING TO MINIMIZE THE TRANSFER OF INVASIVE AQUATIC SPECIESMEPC 80/17/Add.1 Annex 17, page 6 - 2023 GUIDELINES FOR THE CONTROL AND MANAGEMENT OF SHIPS' BIOFOULING TO MINIMIZE THE TRANSFER OF INVASIVE AQUATIC SPECIESMEPC 80/17/Add.1 Annex 17, page 6 I:\MEPC\80\MEPC 80-17-Add.1.docx Figure 1: Simplified flow chart visualizing the biofouling management activities of a ship RESOLUTION MEPC.378(80) (adopted on 7 July 2023) 2023 GUIDELINES FOR THE CONTROL AND MANAGEMENT OF SHIPS' BIOFOULING TO MINIMIZE THE TRANSFER OF INVASIVE AQUATIC SPECIESMEPC 80/17/Add.1 Annex 17, page 7 I:\MEPC\80\MEPC 80-17-Add.1.docx 2 DEFINITIONS 2.1 For the purposes of these Guidelines, the following definitions apply: Anti-fouling system (AFS) means a coating, paint, surface treatment, surface or device that is used on a ship to control or prevent attachment of organisms.

Anti-fouling coating (AFC) means a surface coating or paint designed to prevent, repel or facilitate the detachment of biofouling from hull and niche areas that are typically or occasionally submerged.

- facilitate the detachment of biofouling from hull and niche areas that are typically or occasionally submerged.

Biofouling is the accumulation of aquatic organisms such as microorganisms, plants and animals on surfaces and structures immersed in or exposed to the aquatic environment.

Biofouling can include pathogens.

For microfouling and macrofouling, see definitions below.

Biofouling pressure means the biofouling accumulation rate , which differs regionally and seasonally.

High biofouling pressure means the development of dense biofouling within a short period of time.

Capture is the process of containment, collection and removal of biofouling material and waste substances detached from submerged surfaces during cleaning in water or in dry dock.

Cleaning system is the equipment used for, or the process of, removal of biofoul